

4.16 use average bond energies to calculate the enthalpy change during a simple chemical reaction

c) Rates of reaction

Students will be assessed on their ability to:

- 4.17 describe experiments to investigate the effects of changes in surface area of a solid, concentration of solutions, temperature and the use of a catalyst on the rate of a reaction
- 4.18 describe the effects of changes in surface area of a solid, concentration of solutions, pressure of gases, temperature and the use of a catalyst on the rate of a reaction
- 4.19 understand the term 'activation energy' and represent it on a reaction profile
- 4.20 explain the effects of changes in surface area of a solid, concentration of solutions, pressure of gases and temperature on the rate of a reaction in terms of particle collision theory
- 4.21 understand that a catalyst speeds up a reaction by providing an alternative pathway with lower activation energy

d) Equilibria

Students will be assessed on their ability to:

- 4.22 recall that some reactions are reversible and are indicated by the symbol $\rightleftharpoons$ in equations
- 4.23 describe reversible reactions such as the dehydration of hydrated copper(II) sulfate and the effect of heat on ammonium chloride
- 4.24 understand the concept of dynamic equilibrium
- 4.25 predict the effects of changing the pressure and temperature on the equilibrium position in reversible reactions